

# Day 26 Student Handout

Geometry Summer Credit | Cumulative Synthesis Review

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| Name: | Date: | Period: |
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## Learning Targets

I can review circles, measurement, coordinate geometry, and modeling. I can decide which formula or strategy fits a mixed problem. I can prepare for the final capstone and written final.

## Lesson 26A: Circles and Measurement Review

### Warm-Up

|    |                                                                           |
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| 1. | Write the equation of a circle with center (2, -5) and radius 6.          |
| 2. | Find the arc length for radius 9 and central angle 80 degrees.            |
| 3. | Find the volume of a cylinder with radius 4 and height 10 in terms of pi. |

## Formula Choice

Circle equation uses center and radius. Arc length uses a fraction of circumference. Sector area uses a fraction of circle area. Surface area covers the outside; volume fills the inside.

| Problem                                                                          | Work / notes                              |
|----------------------------------------------------------------------------------|-------------------------------------------|
| A circle has equation $(x + 3)^2 + (y - 4)^2 = 49$ . Name the center and radius. | Center:<br>Radius:                        |
| A circle has radius 12 and central angle 150 degrees. Find sector area.          | Fraction:<br>Circle area:<br>Sector area: |
| A rectangular prism is 8 by 5 by 6. Find surface area and volume.                | Surface area:<br>Volume:<br>Units:        |

# Day 26 Student Handout

Lesson 26B | Coordinate Geometry and Modeling Review

## Coordinate and Modeling Tools

### Strategy Reminders

Use slope for parallel/perpendicular evidence, distance for length/radius, midpoint for centers, and units to decide whether a modeling problem asks for length, area, volume, mass, or density.

| Problem                                                                                                                                           | Work / notes                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Find the distance and midpoint between A(-4, 7) and B(8, -2).                                                                                     | Distance:<br>Midpoint:          |
| A package can weigh no more than 1800 g. Its volume is $250 \text{ cm}^3$ and density is $6 \text{ g/cm}^3$ . Does it meet the weight constraint? | Mass:<br>Compare:<br>Decision:  |
| Two similar solids have scale factor 3. The smaller volume is 35 cubic units. Find the larger volume.                                             | Volume scale:<br>Larger volume: |

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| 4. | Name one topic you feel confident about and one topic you want to review before the final. |
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| 5. | Explain how units help you choose between area, surface area, volume, and density. |
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